

Chinthakommadinne Vinay Kumar

+91 6301077838 | cvinay2108@gmail.com | Github: vinay-0821 | LinkedIn: cvinay2108

Education

National Institute of Technology, Karnataka, Surathkal

Aug. 2025 – Present

M.Tech. Computer Science and Engineering – Information Security | GPA: 8.86/10

National Institute of Technology, Calicut

Dec. 2021 – May 2025

B.Tech. Computer Science and Engineering | GPA: 8.31/10

Experience

Data & AI Intern

May 2026 – July 2026

Accenture

- Developed an open-source video indexing pipeline replacing Azure AI Video Indexer, eliminating recurring cloud API costs using **FFmpeg**, **Faster-Whisper**, and **Ollama** for metadata extraction, transcription, summarization, keyword extraction, and sentiment analysis (reduced processing time by 40% through parallel execution).
- Integrated EasyOCR and YOLO-based object detection to enrich video metadata with on-screen text and detected objects, enabling multimodal indexing and intelligent video search.
- Built a hybrid **RAG** retrieval system using BM25 and Sentence-Transformers for keyword and semantic video search, and developed an AI-powered smart clip recommendation and export pipeline supporting **16:9** and **9:16** downloadable formats.

Software Development Engineer Intern

May 2024 – July 2024

Yume

- Developed a responsive React web application and integrated **AWS Amplify** for secure sign-up, login.
- Deployed the application on AWS Amplify Hosting with CI/CD pipelines for automated build and deployment on every code push.

Projects

Online BookStore Application

GitHub

- Developed a full-stack online bookstore application using React (TypeScript), Node.js (Express), MySQL, and Redux Toolkit, with **JWT-based authentication**, secure session management, and role-based access control across **3 roles**(Customer, Seller, Admin).
- Designed **20+ RESTful APIs** for user, book, and order management, and built an Admin Dashboard to visualize sales and user analytics.

Multithreaded File Downloader

GitHub

- Built a **multithreaded HTTP/HTTPS file downloader** in **C++17** using **libcurl** and a custom **thread pool**, splitting downloads into parallel **byte-range (HTTP Range) requests** to maximize throughput on large files.
- Engineered **thread-safe, zero-copy disk writes** via **pwrite** into a preallocated output file, and implemented **exponential backoff retry logic** with fault isolation, along with a live multi-bar **progress tracker** and structured logging for reliability.

Augmented Reality for Primary Education

GitHub

- Developed an AR educational application that displays interactive 3D models for English alphabets (A–Z) to enhance early literacy.
- Built the vision pipeline by training a custom **YOLOv8m** model (**mAP ≥ 0.85**), integrating it with **Unity Sentis** via **ONNX** for real-time inference and **Google MediaPipe** for gesture control.

Technical Skills

Languages: C, C++, Java, Python, JavaScript, SQL

AI Technologies: Machine learning, AI, Gen AI, Agentic AI, RAG, LLMs, Prompt Engineering

Web Development: HTML5, CSS3, React.js, Node.js, Express.js, Spring Boot, REST APIs, JWT Authentication

Tools: Git, GitHub, Docker, Linux, Claude Code, Github Copilot

Courseworks: DSA, OOP, DBMS, Operating Systems, Computer Architecture, Computer Networks, Network Security, Database Security

Achievements

- AIR 5023 in JEE Main | AIR 8892 in JEE Advanced | AIR 2498 in GATE CS & IT